

optimization of multistage Rocket:

$$\Delta u_{b,imp} = \sum_{i=1}^n \underbrace{g_e}_{\text{known}} \underbrace{I_{sp_i}}_{\text{Fixed}} \ln \left(\underbrace{\frac{1+\lambda_i}{\epsilon_i+\lambda_i}}_{\text{Fixed}} \right)$$

fixed for maximization of $\frac{M_L}{M_{01}}$

optimize λ_i

$$\lambda_i = \frac{M_{Li}}{M_0 - M_{Li}}$$

Variation $\leftarrow \delta(\Delta u_{b,imp}) = 0$

$$\delta \left[\sum_{i=1}^n g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \right] = 0$$

$$\Rightarrow \sum_{i=1}^n \delta \left[g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \right] = 0$$

$$\delta = \frac{\partial}{\partial \lambda_i} d\lambda_i \Rightarrow \sum_{i=1}^n \frac{d}{d\lambda_i} \left[g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \right] d\lambda_i = 0$$

$$\sum_{i=1}^n g_e I_{sp_i} \frac{\epsilon_i + \lambda_i}{1 + \lambda_i} \left[\frac{\epsilon_i + \lambda_i - (1 + \lambda_i)}{(\epsilon_i + \lambda_i)^2} \right] d\lambda_i = 0$$

$$\therefore \sum_{i=1}^n g_e I_{sp_i} \frac{\epsilon_i - 1}{(1 + \lambda_i)(\epsilon_i + \lambda_i)} \delta \lambda_i = 0$$

(A)

$$\delta(\Delta u_{b,imp}) = 0$$

$$\frac{\partial}{\partial \lambda_i} [(*)] d\lambda_i = 0$$

$$\frac{M_L}{M_{01}} = \prod_{i=1}^n \left(\frac{\lambda_i}{1+\lambda_i} \right)$$

$$\delta \left(\frac{M_L}{M_{01}} \right) = 0 \quad \text{Second constraint}$$

$$\Rightarrow \delta \left[\prod_{i=1}^n \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

(or)

$$\sum_{i=1}^n \delta \left[\ln \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

$$\sum_{i=1}^n \frac{1}{\lambda_i (1 + \lambda_i)} \delta \lambda_i = 0$$

(B)

B

if we use Lagrange multiplier for (A) + (B) then

$$\textcircled{B} + \alpha \textcircled{A} = 0 \text{ to get } \underset{\text{minima}}{\text{Max}} \lambda_i$$

$$\underbrace{\sum_{i=1}^n \frac{1}{\lambda_i (1+\lambda_i)} \delta \lambda_i}_B + \alpha \underbrace{\left(\sum_{i=1}^n g_e I_{SP_i} \frac{\epsilon_i - 1}{(1+\lambda_i)(\epsilon_i + \lambda_i)} \delta \lambda_i \right)}_A = 0$$

Here λ_i is independent

$$\boxed{\frac{1}{\lambda_i (1+\lambda_i)} + \alpha g_e I_{SP_i} \frac{\epsilon_i - 1}{(1+\lambda_i)(\epsilon_i + \lambda_i)}} = 0 \rightarrow \textcircled{C}$$

$$\frac{1}{\lambda_i (1+\lambda_i)} = \frac{(1-\epsilon_i) \alpha g_e I_{SP_i}}{(1+\lambda_i)(\epsilon_i + \lambda_i)}$$

$$\frac{\epsilon_i + \lambda_i}{\lambda_i} = (1-\epsilon_i) \alpha I_{SP_i} g_e$$

$$\frac{\epsilon_i}{\lambda_i} + 1 = \alpha I_{SP_i} g_e (1-\epsilon_i)$$

$$\frac{\epsilon_i}{\lambda_i} = \underbrace{\alpha I_{SP_i} g_e (1-\epsilon_i)}_{\text{known}} - 1$$

$$\boxed{\lambda_i = \frac{\epsilon_i}{\underbrace{\alpha g_e I_{SP_i} (1-\epsilon_i)}_{\text{unknown}} - 1}} \rightarrow \textcircled{D}$$

' α ' is obtained from the eq $\Delta U_{b,imp}$

$$\boxed{\Delta U_{b,imp} = \sum_{i=1}^n g_e I_{SP_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i + \lambda_i} \right)}$$

$\downarrow$ fixed $\downarrow$ known $\downarrow$ fixed $\downarrow$ fixed

$$\Delta U_{b,imp} = \sum_{i=1}^n g_e I_{SP_i} \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}} \right]$$

λ_i

$$1 = \sum_{i=1}^n \left[\frac{g_e I_{SP_i}}{\Delta U_{b,imp}} \right] \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}} \right]$$

$$1 = \sum \square \ln O$$

$$1 = \sum \ln O \quad \square$$

$$e^1 = e^{\sum \ln O} \quad \square$$

$$1 = \sum_{i=1}^n \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}} \right] \left[\frac{g_e I_{SP_i}}{\Delta U_{b,imp}} \right]$$

$$e = e^{\sum_{i=1}^n \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}} \right] \left[\frac{g_e I_{SP_i}}{\Delta U_{b,imp}} \right]}$$

$$e^{\sum_{i=1}^n \ln O} = \prod_{i=1}^n O \quad \square$$

$$e = \prod_{i=1}^n \left(\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1}} \right)^{\frac{g_e I_{SP_i}}{\Delta U_{b,imp}}}$$

Simplified

$$e = \prod_{i=1}^n \left(\frac{\alpha g_e I_{SP_i} (1 - \epsilon_i) - 1 + \epsilon_i}{\epsilon_i \alpha g_e I_{SP_i} (1 - \epsilon_i) - \cancel{\epsilon_i} + \cancel{\epsilon_i}} \right)^{\frac{g_e I_{SP_i}}{\Delta U_{b,imp}}}$$

$$e = \prod_{i=1}^n \left(\frac{\cancel{\alpha g_e I_{SP_i} (1 - \epsilon_i)} - 1 + \epsilon_i}{\epsilon_i \cancel{\alpha g_e I_{SP_i} (1 - \epsilon_i)}} \right)^{\frac{g_e I_{SP_i}}{\Delta U_{b,imp}}}$$

$$= \prod_{i=1}^n \left[\frac{1}{\epsilon_i} - \frac{(1-\epsilon_i)}{\epsilon_i \alpha g_e I_{sp_i} (1-\epsilon_i)} \right] \frac{g_e I_{sp_i}}{\Delta U_{b,i} \eta_p}$$

Neper No.
2.71 = e =

$$\prod_{i=1}^n \left[\frac{1}{\epsilon_i} - \frac{1}{\epsilon_i \alpha g_e I_{sp_i}} \right] \frac{g_e I_{sp_i}}{\Delta U_{b,i} \eta_p}$$

$$\therefore \left[\frac{1}{\epsilon_1} - \frac{1}{\epsilon_1 \alpha g_e I_{sp_1}} \right] \frac{g_e I_{sp_1}}{\Delta U_{b,i} \eta_p} \times \left[\frac{1}{\epsilon_2} - \frac{1}{\epsilon_2 \alpha g_e I_{sp_2}} \right] \frac{g_e I_{sp_2}}{\Delta U_{b,i} \eta_p} \dots \dots \dots \left[\frac{1}{\epsilon_n} - \frac{1}{\epsilon_n \alpha g_e I_{sp_n}} \right] \frac{g_e I_{sp_n}}{\Delta U_{b,i} \eta_p} \Bigg\} = e = 2.718$$

Solve this & get α

implicit, non-linear Eqⁿ
for α

if $\alpha = \frac{2+n}{3}$ ^{known} : explicit, linear

Example of optimization

Example

Reconsider the earlier three-stage rocket example.

$$\widehat{I_{sp1}} = \widehat{I_{sp2}} = \widehat{I_{sp3}} = 370 \text{ s}$$

$$\widehat{\epsilon_1} = 0.05, \widehat{\epsilon_2} = 0.0698, \widehat{\epsilon_3} = 0.1892$$

$$f_{\text{fixed}} = f\left(\frac{M_L}{M_{01}}\right) = \frac{68,000}{2,500,000} = 0.0272$$

$$\text{With } \lambda_1 = 0.3021, \lambda_2 = 0.4464, \lambda_3 = 0.6126,$$

$$\Delta u_{b, \text{imp}} = 11023 \text{ m/s.}$$

Let us optimize the vehicle to maximize $\left(\frac{M_L}{M_{01}}\right)$

for $\Delta u_{b, \text{imp}} = 11023 \text{ m/s}$ fixed

$$\left\{ \frac{1}{\epsilon_1} \left[1 - \frac{1}{\alpha g_e I_{sp1}} \right] \right\}^{\frac{g_e I_{sp1}}{\Delta u_{b, \text{imp}}}} \otimes \left\{ \frac{1}{\epsilon_2} \left[1 - \frac{1}{\alpha g_e I_{sp2}} \right] \right\}^{\frac{g_e I_{sp2}}{\Delta u_{b, \text{imp}}}} \otimes \left\{ \frac{1}{\epsilon_3} \left[1 - \frac{1}{\alpha g_e I_{sp3}} \right] \right\}^{\frac{g_e I_{sp3}}{\Delta u_{b, \text{imp}}}} = e^{\textcircled{E}}$$

USE MATLAB to solve Equation $\textcircled{E}$ for αg_e

Then, get λ_i from Eqn. $\textcircled{D}$:

$$\lambda_i = \frac{\epsilon_i}{\alpha g_e I_{spi} (1 - \epsilon_i) - 1}$$

$$\frac{M_L}{M_{01}} = \frac{\lambda_1}{1 + \lambda_1} \times \frac{\lambda_2}{1 + \lambda_2} \times \frac{\lambda_3}{1 + \lambda_3} \quad [\text{from (10.35)}]$$

known from
example

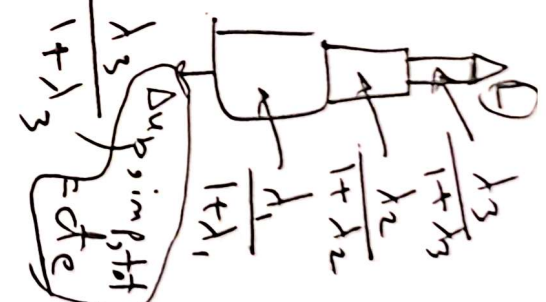
MATLAB solution:

$$\alpha q_2 = 0.003554$$

$$\lambda_1 = 0.2005, \lambda_2 = 0.3125, \lambda_3 = 2.8524$$

$$\left(\frac{M_L}{M_{01}}\right)_{opt} = 0.0294$$

$$\left(\frac{M_L}{M_{01}}\right)_{opt} = \left(\frac{\lambda_1}{1+\lambda_1} \times \frac{\lambda_2}{1+\lambda_2} \times \frac{\lambda_3}{1+\lambda_3}\right)$$



$$\Delta v_{b,imp1} = 5688.1 \text{ m/s}, \Delta v_{b,imp2} = 4477.2 \text{ m/s}$$

$$\Delta v_{b,imp3} = 857.7 \text{ m/s}$$

$$\Delta v_{b,imp} = \sum_{i=1}^3 \Delta v_{b,imp i} = 11023 \text{ m/s} = cte \text{ as before}$$

* If M_L is maintained at 68000 kg, (as given in example)

$$(M_{01})_{opt} = \frac{M_L}{(M_L/M_{01})_{opt}} = \frac{68000}{0.0294} \approx 2,313,000 \text{ kg} \cdot 2.5 \text{ Mkg}$$

Thus the optimized design is capable of imparting the

Δv_b of 11023 m/s to the 68000 kg payload with a lighter

launch vehicle [2,313,000 kg vs 2,500,000 kg]

* If, on the other hand, M_{01} is maintained at 2,500,000 kg,

$$(M_L)_{opt} = M_{01} \left(\frac{M_L}{M_{01}}\right)_{opt} = 2,500,000 (0.0294) = 73500 \text{ kg}$$

Thus the optimized 2,500,000 kg launch vehicle is capable of providing the Δv_b of 11023 m/s to a more

massive payload (73650 kg vs 68000 kg)

$$M_{Lopt} > M_L$$

Example of optimization (Equal weight)

Example

Consider a 3-stage rocket.

$$\epsilon_1 = 0.0765, \epsilon_2 = 0.114, \epsilon_3 = 0.111$$

$$I_{sp1} = 304.5, I_{sp2} = 421.5, I_{sp3} = 421.5$$

$$\Delta u_{b,imp} = 1204 \text{ km/s}$$

optimize the vehicle for maximum $\left(\frac{M_L}{M_{01}}\right)$.

$$\left\{ \frac{1}{\epsilon_1} \left[1 - \frac{1}{\alpha g_e I_{sp1}} \right] \right\}^{\frac{g_e I_{sp1}}{\Delta u_{b,imp}}} \otimes \left\{ \frac{1}{\epsilon_2} \left[1 - \frac{1}{\alpha g_e I_{sp2}} \right] \right\}^{\frac{g_e I_{sp2}}{\Delta u_{b,imp}}} \otimes \left\{ \frac{1}{\epsilon_3} \left[1 - \frac{1}{\alpha g_e I_{sp3}} \right] \right\}^{\frac{g_e I_{sp3}}{\Delta u_{b,imp}}} = e \quad (E)$$

α

Use MATLAB to solve Equation (E) for αg_e

Then, get λ_i from Eqn. (D):

$$\lambda_i = \frac{\epsilon_i}{\alpha g_e I_{sp_i} (1 - \epsilon_i) - 1}$$

put α in
 $\lambda_i = f(\epsilon_i, I_{sp_i})$

$$\frac{M_L}{M_{01}} = \frac{\lambda_1}{1 + \lambda_1} \cdot \frac{\lambda_2}{1 + \lambda_2} \cdot \frac{\lambda_3}{1 + \lambda_3} \quad [\text{from (10.35)}]$$

MATLAB solution:

$$\alpha g_e = 0.0038966$$

$$\lambda_1 = 0.8142, \lambda_2 = 0.2514, \lambda_3 = 0.2421$$

$$\frac{M_L}{M_{01}} = 0.01758$$

$$\Delta u_{b, \text{imp} i} = \overset{= I_{sp} g_e}{u_{eq i}} \ln \left(\frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \right)$$

$$(\Delta u_{b, \text{imp} 1} = 2121.5 \text{ m/s}, \Delta u_{b, \text{imp} 2} = 5084.2 \text{ m/s},$$

$$\Delta u_{b, \text{imp} 3} = 5194.3 \text{ m/s})$$

$$\Delta u_{b, \text{imp}} = \sum_{i=1}^3 \Delta u_{b, \text{imp} i} = 12400 \text{ m/s})$$

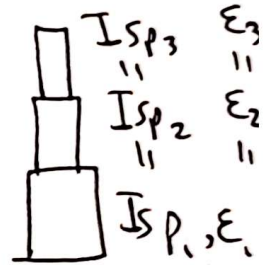
The data of this problem correspond to the

Saturn V lunar launch vehicle. Its initial mass

(M_{01}) was 2,902,000 kg and the mass of the Apollo 11

payload (M_L) was 47000 kg, which gives $\frac{M_L}{M_{01}} = 0.0162$.

A vehicle with stages having equal ϵ_i and $I_{sp i}$ values is said to be Similarly Staged. It is



clear from Eqn (D), that the stages will have equal values of λ_i as well. For such a vehicle, Eqn. (E)

becomes
$$\Delta u_{b, \text{imp}} = \sum_{i=1}^n I_{sp i} g_e \ln \left(\frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \right) = \sum_{i=1}^n 0 = n \cdot 0$$

$$\textcircled{n} g_e I_{sp} \ln \left(\frac{1 + \lambda}{\epsilon + \lambda} \right) = \Delta u_{b, \text{imp}} \textcircled{F}$$

where n is the number of stages, and

$$\overset{\text{def}}{I_{sp}} = I_{sp 1} = I_{sp 2} = I_{sp 3},$$

$$\overset{\text{def}}{\lambda} = \lambda_1 = \lambda_2 = \lambda_3, \text{ and}$$

$$\overset{\text{def}}{\epsilon} = \epsilon_1 = \epsilon_2 = \epsilon_3.$$

known in
optimization

In this case, λ can be found from Eqn (F), without

having to find α . If needed, α can be found from
 Eqn (D) $\left\{ \lambda_i = \frac{\epsilon_i}{\alpha g_e I_{sp_i} (1 - \epsilon_i) - 1} \right\}$ once λ has been
 determined.

Example

Reconsider the first three-stage example $\{M_{01} = 2,500,000 \text{ kg}, \Delta u_b = 11023 \text{ m/s}\}$. The stages all have the same I_{sp} .

$M_s = 145,000 \text{ kg}, M_p = 2,287,000 \text{ kg}$. If the structural and propellant masses are redistributed such that $\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon$, $\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon$

$$\epsilon = \frac{M_{s1}}{M_{s1} + M_{p1}} = \frac{M_{s2}}{M_{s2} + M_{p2}} = \frac{M_{s3}}{M_{s3} + M_{p3}} =$$

$$\frac{M_{s1} + M_{s2} + M_{s3}}{(M_{s1} + M_{p1}) + (M_{s2} + M_{p2}) + (M_{s3} + M_{p3})} = \frac{M_{s\text{tot}}}{M_{s\text{tot}} + M_{p\text{tot}}} =$$

for this
rocket

$$\frac{M_L}{M_0} = \prod_1^n \frac{\lambda_i}{1 + \lambda_i} = \left(\frac{\lambda}{1 + \lambda} \right)^n$$

$$\frac{145000}{145000 + 2287000} \approx 5\% = 0.05962$$

Solve

With this value of ϵ , $\textcircled{F} \Rightarrow \lambda = 0.4772$ [$n = 3$]

$$\frac{M_L}{M_{01}} = \left(\frac{\lambda}{1 + \lambda} \right)^{\textcircled{3}} = 0.0337 \quad \text{larger than the } 0.0294 \text{ value}$$

obtained with $\epsilon_1 = 0.05, \epsilon_2 = 0.0698$ and $\epsilon_3 = 0.1892$

Previous case

The value of αg_e (not really needed) can be found from (D) :

$$\alpha g_e = \frac{1}{I_{sp}(1-\epsilon)} \left(1 + \frac{\epsilon}{\lambda} \right) = \underline{0.03233}$$

The similarly-staged vehicle has the highest $\left(\frac{M_2}{M_{01}} \right)$.